Financial Modeling

Class 3 — Portfolio Construction

In-class session

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Where we are in the course

- The three blocks of the course: **evaluation**, **construction**, **strategies**
- So far: markets are efficiently inefficient (Class 1); performance can be measured, but separating skill from noise requires decades of data (Class 2)
- Today begins the **construction** block, which occupies three sessions: how an institutional investor builds a portfolio
- This block prepares AMC round 1, the optimization round

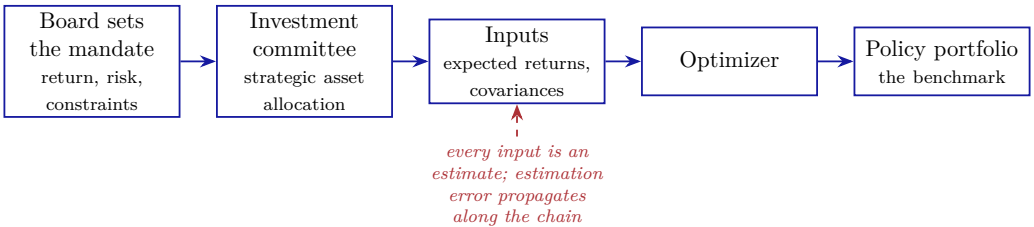
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Today: the allocation decision

- An institution sets the weights across asset classes and risk premia before any security is selected. This decision determines most of the risk and return of the portfolio
- Markowitz (1952) provides the formal answer: an optimizer that maps estimates of means and covariances into weights
- The question for today is how the optimizer behaves when its inputs are estimated with error
- The answer explains how institutions allocate in practice, and it applies to the AMC portfolios

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From mandate to policy portfolio



This is the allocation process of an institutional investor. The class follows it from the mandate to the policy portfolio.

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Exercise: the allocation of a pension fund

- You are the CIO of a **CHF 2 billion** pension fund in Ticino. The board sets:
  - a volatility budget of **6%**
  - **no short positions**, leverage at most **10%**
  - **CHF 80 million** a year to pay out in pensions
- The investable universe: **global equities, government bonds, illiquid alternatives (PE + hedge funds), alternative risk premia, cash**
- **In pairs, take five minutes to write down the portfolio weights in the Strategic Asset Allocation.** They must sum to 100%.

Three of the portfolios are written on the board. We return to them at the end of the class.

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The four macro regimes, from the reading

- Growth Up + Inflation Up: **Overheated economy**
- Growth Up + Inflation Down: **Goldilocks**
- Growth Down + Inflation Up: **Stagflation**
- Growth Down + Inflation Down: **Depression**

Before looking at the data,

we vote on each question on Wooclap.

Open the event now, from your phone: `wooclap.com/YDPSCLL`



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Vote 1

In stagflation, the Sharpe Ratio of global equities is roughly:

$+0.3$ 0 -0.3

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Vote 2

In which regime do government bonds earn their best Sharpe Ratio?

Overheated Goldilocks Stagflation Depression

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Vote 3

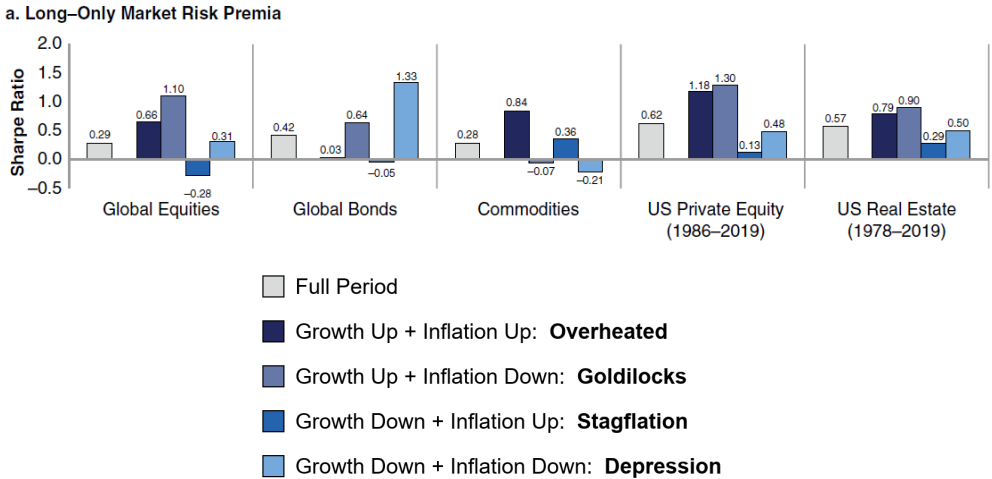
Which traditional asset class is the only one that performs well under inflation?

Equities Bonds Commodities

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The data: traditional asset classes by regime



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Vote 4

The Sharpe Ratio of equities changes sign across regimes ($+1.10$ to -0.28).

Does the Sharpe Ratio of momentum change sign as well?

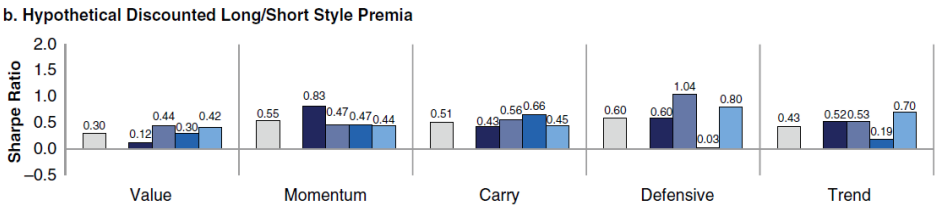
Yes No

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The data: style premia by regime



- The Sharpe Ratios lie between 0.44 and 0.83 in every regime; none is negative
- (Sharpe Ratios discounted for backtesting bias and trading costs)

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Vote 5

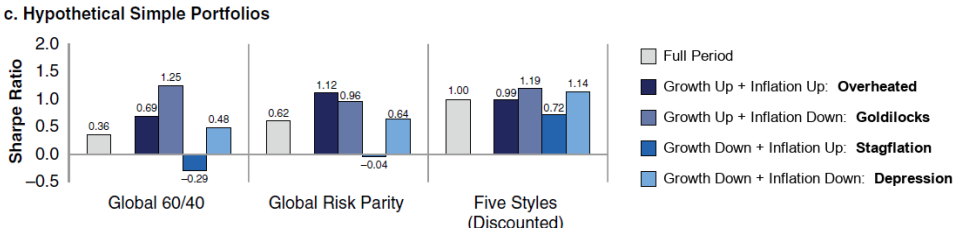
Which portfolio keeps a positive Sharpe Ratio in *every* regime?

60/40 Risk parity Basket of five styles

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The data: three simple portfolios by regime



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What the chart shows

- Traditional asset classes are **exposures to specific regimes**: equities perform in Goldilocks, bonds in Depression, commodities under inflation
- Style premia are **stable across regimes**
- An investor who could predict regimes would rotate across asset classes; this is Tactical Asset Allocation.
  Predicting regimes is difficult
- The Strategic Asset Allocation should therefore be designed to **perform acceptably in every regime**

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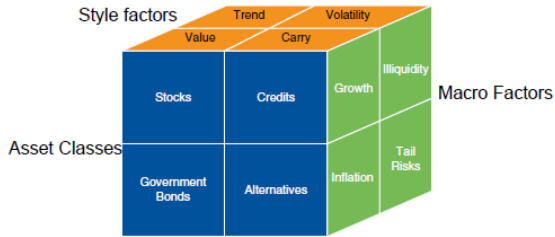
Recap 1 of 4: the layers of the allocation decision

1. **Strategic Asset Allocation** — the policy portfolio, which serves as the benchmark: the long-run mix of the **traditional betas** (equity, term, and credit premia)
2. **Factor/style tilts** — long-run exposure to the **alternative betas** (value, momentum, carry, defensive), often inside the SAA
3. **Tactical Asset Allocation** — medium-term deviations
4. **Security selection** — "alpha"

- Brinson, Hood, and Beebower: **the policy portfolio drives the large majority of the variation in returns**
- The SAA is the first decision to get right: it carries most of the risk

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Recap 2 of 4: asset classes and factors



- Every asset class is a **combination of factor exposures**. Asset classes with different names can depend on the same factor: equities, credit, and private equity all depend on growth. A portfolio diversified by asset class can be concentrated in one factor
- Asset classes and styles are investable; macro factors are reached through them

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Recap 3 of 4: what the fees pay for

- **Market beta** (the traditional betas) — cheap, scalable, and the largest part of the risk; it should be bought passively
- **Alternative beta** (styles, rule-based, long–short) — moderate fees
- **Alpha** — scarce, expensive, hard to identify in advance

Fees should be proportional to scarcity.
Alpha fees should not be paid for repackaged beta ("closet indexing").

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Recap 4 of 4: what the optimizer does

- **Risky assets only.** Minimize the variance for a given expected return m, with weights that sum to one: $\min_x x'Vx$ s.t. $x'\mu = m,\ x'\iota = 1$.
  The solutions, one for each m, form a hyperbola

- **With a risk-free asset.** The weights of the risky assets need not sum to one: the rest, $1 - x'\iota$, is in the risk-free asset. With $\mu^e = \mu - r_f$,

$$\min_x\ x'Vx \quad \text{s.t. } x'\mu^e = e \qquad \Longrightarrow \qquad x = \lambda\, V^{-1}\mu^e$$

  The risky assets are held in the same proportions for every e: the frontier is a straight line

- The portfolio on this line with $x'\iota = 1$ is the **tangency portfolio**, which has the **maximum Sharpe Ratio**: $x_T = \dfrac{V^{-1}\mu^e}{\iota' V^{-1}\mu^e} = \arg\max_x \dfrac{x'\mu^e}{\sqrt{x'Vx}}$

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In Excel, round 1: two assets

	$E(R^e)$	σ
Equities	5%	15%
Bonds	1%	5%

Correlation: zero

Does the portfolio with the highest Sharpe Ratio hold more or less than half of the capital in equities?

more than half less than half

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Round 1: why these weights?

With zero correlation, the maximum Sharpe Ratio requires $w_i \propto \frac{E(R_i^e)}{\sigma_i^2}$

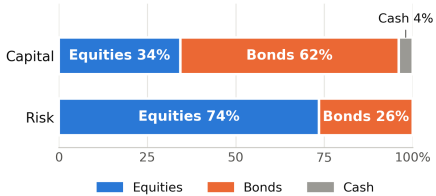
	$E(R^e)$	σ	$E(R^e)/\sigma^2$	Capital weight w	$w \times \sigma$	Share of the risk
Equities	5%	15%	2.22	**36%**	5.4%	**74%**
Bonds	1%	5%	4.00	**64%**	3.2%	**26%**

- **Capital.** Equities have 5 times the expected return of bonds, but 9 times the variance $\Rightarrow$ bonds receive more capital
- **Risk.** A franc in equities is three times as volatile as a franc in bonds. Equities move the portfolio by $36\% \times 15\% = 5.4\%$, bonds by $64\% \times 5\% = 3.2\%$. In variance, equities are $5.4^2/(5.4^2 + 3.2^2) = 74\%$
- By capital the portfolio is 64% bonds. By risk it is 74% equities

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Round 1 and the pension fund

The mandate of slide 4, with equities and bonds only: volatility **6%**, no short positions, leverage at most 10%.



Expected excess return	2.3%
Volatility	6.0%
Sharpe Ratio	0.39

- This is the mix of round 1 plus 4% in cash. The limit on leverage is not binding
- Can the fund pay the pensions? They are CHF 80 million a year, **4%** of the assets. The portfolio is expected to earn the risk-free rate plus **2.3%** (CHF 47 million). It covers the pensions only if the risk-free rate is at least 1.7%
- Compare with the portfolios on the board

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In Excel, round 2: four assets

Add Illiquid Alternatives and Alternative Risk Premia:

	$E(R^e)$	σ	Sharpe Ratio
Equities	5%	15%	0.33
Bonds	1%	5%	0.20
Illiquid Alternatives	7%	10%	0.70
Alternative Risk Premia	7%	10%	0.70

Correlation between equities and illiquid alternatives: **0.70**. Two assets now have twice the Sharpe Ratio of equities.

How does the optimal allocation change?

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Round 2: summary of the results

Inputs	Equities	Bonds	Illiq. Alts	ARP	SR
As estimated	$-$**20**%	-7%	+73%	+53%	0.98
Equity premium 5% $\to$ 6%	-12%	-4%	+64%	+52%	0.96
Equity premium 5% $\to$ 4%	-28%	-11%	+83%	+55%	1.01
Illiquid Alts 7% $\to$ 6%	-12%	0%	+57%	+55%	0.89

- With 50 years of data on a 15%-volatility asset, the standard error of the mean is $15\%/\sqrt{50} \approx$ **2.1 percentage points**
- The $\pm$1pp changes above are *within the estimation error*, and they move the equity weight from -12% to -28%

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According to the spreadsheet, the optimal portfolio for the pension fund is

short CHF 400 million of equities.

1. **Why does the optimizer short equities?**
2. **Would you propose it to the board?**

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Interpreting the result

The optimizer did what it was asked to do.

For which reasons should the CIO not rely on its output?

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The two sources of error

- The short equity position is **consistent with the inputs**: with a correlation of 0.70, shorting equities hedges the position in illiquid alternatives. The optimizer works as intended; **the problem lies in the model and in its inputs:**
  - **Model error** — mean–variance ignores illiquidity, shorting limits, and the fact that appraisal-smoothed data understate the risk of the alternatives
  - **Estimation error** — the weights change substantially in response to input changes smaller than the estimation error
- **Constraints do not solve it**: under the fund's mandate (6%, no short positions, leverage at most 10%) the optimizer holds no equities, 1% bonds, 40% illiquid alternatives, 40% ARP, 18% cash. The short position is gone, but the inputs are the same and the portfolio is still extreme

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Back to your portfolios

Each pair revises **at most two numbers**
and states what the revision addresses:

the mandate **model error** **estimation error**

I will then ask three pairs to present their revised weights and their reasoning.

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Assignment: your investment policy (AMC checkpoint 1)

In round 1 of the AMC each team chooses its **Strategic Asset Allocation**. One page per group, the **first page of your round-1 report**:

1. **Universe** — the broad-asset-class ETFs that the allocation may hold. Round 1: **long-only ETFs**, no leverage, few positions, cash earns no interest
2. **Objective and risk budget** — the evaluation is on the **Sharpe Ratio**, and the guidelines of the round set the **volatility target**. State the ex ante volatility of your allocation, the corresponding Value at Risk, and the maximum acceptable drawdown
3. **Constraints, with reasons** — beyond the rules of the round: concentration, regions, liquidity, each with its reason in terms of today's model errors
4. **Risk contributions** — for your weights: capital weight *and* risk contribution of each position. State whether a position above half of the risk is intended

Graded as complete or incomplete, together with the round-1 report.

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Next class

- The construction rules that allocators use when expected returns cannot be estimated reliably:
  - **minimum variance**
  - **equal weight**
  - **value weight**
  - **risk parity**
- A horse race among these rules on 40 years of data
- Two results of today motivate them:
  - Expected returns are estimated with error, and the optimizer's weights are unreliable (Excel). These rules **do not use expected returns**
  - 60/40 has 90% of its risk in equities: Sharpe Ratio 0.36, and -0.29 in stagflation. Risk parity spreads the risk across asset classes: 0.62, and -0.04 in stagflation (Vote 5). Some of these rules **balance the risk**

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